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**Pakistan Studies SSC-II
(9th Book NBF)**

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Chapter # 5

SHORT QUESTION AND

ANSWER:-

Question No. 1:- What are four climatic Zones of Pakistan & what are two key difference between each Zone?

Answer:-

The Climate of Pakistan is broadly divided into four climate Zone due to diverse topographical region.

- * Highland climate.
- * Lowland Climate.
- * Desert or Arid Climate.
- * Coastal or Maritime Climate.

The northern, north western & western mountains experienced a high land climate while the Indus Plain has a Lowland climate.

The Makran Coast, Karachi Coast & Indus Delta experience a Coastal climate while the Southern ~~Stem~~ part of Balochistan have a desert or arid climate.

Question No. 2:- Describe the characteristics of the desert or ~~arid~~ arid climate of Pakistan?

Answer:-

Rainfall

Aridity refers to low precipitation & low moisture content in the environment. The desert areas experience dry & hot Summers & receive scanty rain fall throughout this year.

During the day, hot & dusty winds blow across the plain from May to September. The diurnal range of temperature in desert area is high.

Dust storms are a permanent feature of desert area in high.

Dust storms are a permanent feature of desert climate that temporarily lower the temperature. Winter are cold with the temperature.

Question No 3:-

Explain how the climate of Pothwar Plateau makes it a challenging area for the farmers.

Answer:-

The Pothwar Plateau which is rainfed, are also called "Barani" Land, receives moderate rainfall, ranging from 500 to 1000 mm per year. This is enough rainfall to support crop production, but farmers have to manage water resourcing during dry years.

From May to September, hot & dusty winds blow across the desert plain. The temperature of desert area is extremely high. Dust storms are regular events. The strong temperature lowers the temperature, bringing cooler winds.

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Question no 4:-

How does the coastal climate of Pakistan differ from other climate zones of Pakistan?

Answer:-

The Coastal climate of Pakistan has a maritime climate with a sea breeze round the year. The proximity to sea results in high humidity particularly between April to September. Very little rainfall is experienced the monsoon season. Some rainfall minimum temperature of 4°C .

Question no 5:-

Why do many rivers flood in Pakistan?

Answer:-

A river flood is a natural disaster that occurs when a river overflows its bank & inundates the surrounding land. Although floods deposit fresh nutrients & spread a layer of alluvium in wetlands & the inland delta & recharge ground water but floods & have massive devastating effects on natural & human resources.

• heavy rainfall

• snowmelt

Question no 6:-

How does the yearly distribution of rainfall & temperature vary between Karachi & Lahore?

Answer:-

The region includes Sindh & the major city of Lahore. Southern Pakistan is the hottest region of Pakistan, with average temperature ranging from 28°C in the summer to 18°C in the winter. The region of Pakistan is home

to the port city of Karachi & the desert region of Balochistan.

According to the Pakistan Meteorological Department, the hottest months are ^{May} and ^{June} with average temperature of 30.5°C & 34°C respectively.

The coldest months are December & January with temperature 12.5°C & 15.5°C respectively.

Question No 7:-

Explain the development of Monsoon winds in Arabian Sea.

Answer:-

The Arabian Sea monsoon rains are caused when an active monsoon winds blow across South east of Pakistan & bring rain to the Coastal areas. The Arabian Sea monsoons provide water for agriculture & replenish groundwater supplies. However the monsoon rains also cause flooding landslides & damage to ^{re} infrastructure sometimes.

02/09/24

UNIT NO: 05 (CLIMATE OF PAKISTAN AND ENVIRONMENTAL HAZARDS)

LONG QUESTIONS

✓ Q1. How can the effectiveness of rainfall for agriculture in Pakistan be improved? Discuss the key challenges and opportunities particularly in Indus Plain.

Ans. Improving the effectiveness of rainfall for agriculture in Pakistan, particularly in the Indus Plain, involves addressing several key challenges while leveraging opportunities.

Challenges

1. **Uneven Distribution of Rainfall:** Rainfall in the Indus Plain is not evenly distributed, leading to waterlogging in some areas and drought in others. This inconsistency impacts crop yields and soil fertility.
2. **Water Management Issues:** Inefficient water management practices, such as outdated irrigation systems, lead to significant water losses. Over-reliance on canal irrigation without proper maintenance exacerbates these issues.
3. **Soil Degradation:** Over-irrigation and poor drainage have led to soil salinity and waterlogging in some areas of the Indus Plain, reducing the land's agricultural productivity.
4. **Climate Change:** Changes in weather patterns, including more frequent droughts and unpredictable monsoon rains, pose a significant risk to agriculture.

Opportunities

1. **Rainwater Harvesting:** Implementing rainwater harvesting techniques can help store ~~excess~~ rainwater during the monsoon season for use during dry periods, ensuring a more reliable water supply for crops.
2. **Modern Irrigation Techniques:** Adopting modern irrigation methods such as drip or sprinkler systems can significantly reduce water wastage and improve water use efficiency in agriculture.
3. **Soil Conservation Practices:** Introducing practices like crop rotation, use of organic fertilizers, and controlled irrigation can help in restoring soil health and preventing degradation.
4. **Drought-Resistant Crops:** Promoting the cultivation of drought-resistant crop varieties can help mitigate the impact of variable rainfall and ensure stable agricultural production.
5. **Public Awareness and Training:** Educating farmers on sustainable water management and farming practices is crucial. Extension services can play a key role in disseminating knowledge about modern agricultural techniques.

Q2. To what extent do storms and droughts cause problems for the people of Pakistan? Also suggest some preventive measures and management strategies to reduce the problems.

Ans. Storms and droughts pose significant challenges for the people of Pakistan, impacting their livelihoods, infrastructure, and overall well-being.

Here's an overview of the extent of these problems and some preventive measures and management strategies:

Extent of Problems

1. **Storms:**

- o **Flooding:** Storms, especially during the monsoon season, can lead to ~~severe~~ flooding, damaging homes, crops, and infrastructure. Coastal areas are particularly vulnerable to tropical cyclones that can cause storm surges and coastal flooding.
 - o **Infrastructure Damage:** High winds and heavy rainfall during storms can damage roads, bridges, and buildings, disrupting transportation and communication networks.
 - o **Economic Losses:** Agricultural losses due to storms can be substantial, affecting food security and the livelihoods of farmers. ^{It} Floods often destroy standing crops, leading to a significant economic ~~loss~~ ^{loss}. Storms & floods can significantly impact food security and farmers' livelihoods by damaging crops.
2. **Droughts:** ^{Significantly} ^{led to} ^{Drought} ^{Lead to}
- o **Water Scarcity:** Prolonged droughts lead to severe water shortages, affecting drinking water supplies and irrigation for agriculture. This is particularly critical in arid and semi-arid regions of Pakistan.
 - o **Agricultural Impact:** Droughts reduce crop yields, leading to food insecurity and loss of income for farmers. Livestock may also suffer due to the lack of water and fodder.
 - o **Migration:** In extreme cases, drought can force people to migrate from rural areas to urban centers in search of work and better living conditions, leading to urban overcrowding and stress on resources.

Preventive Measures and Management Strategies

1. **Improved Water Management:**
 - o Implement rainwater harvesting and efficient irrigation techniques such as drip and sprinkler systems to ~~conserve~~ ^{prevent} water.
 - o Develop and maintain reservoirs and dams to store ~~water~~ ^{water} during the rainy season for use during droughts.
2. **Disaster Preparedness:**
 - o Establish early warning systems for storms and droughts to alert communities in advance, allowing them to take precautionary measures.
 - o Develop and implement community-based disaster management plans, including evacuation routes and shelters in storm-prone areas.
3. **Reforestation and Afforestation:**
 - o ^{logs} ~~promote~~ ^{stand with the} reforestation and afforestation ~~programs~~ ^{the} to reduce the impact of storms and prevent soil ~~erosion~~ ^{loss}. Forests act as natural barriers, reducing the velocity of storm winds and stabilizing soil.
4. **Crop Diversification and Drought-Resistant Varieties:**
 - o Encourage farmers to diversify crops and adopt drought-resistant varieties that can withstand dry conditions, ensuring more stable agricultural production.
 - o Provide training and resources to farmers on sustainable agricultural practices that reduce vulnerability to climate extremes.
5. **Infrastructure Resilience:**
 - o Invest in building and upgrading infrastructure to ~~withstand the impacts of~~ ^{stand with the} storms, including flood defenses, storm-resistant buildings, ~~and resilient~~ ^{the} roads and bridges.
 - o Regularly maintain ~~and~~ ^{the} repair existing infrastructure to ~~minimize~~ ^{Reduce} damage during extreme weather events.
6. **Public Awareness and Education:**

Droughts reduce crop yields & harm livestock, leading to food insecurity, income loss for farmers & ~~and~~ increased risk of animal deaths.

- Increase public awareness about the risks associated with storms and droughts and educate communities on how to prepare and respond effectively.
- Engage local communities in environmental conservation efforts to mitigate the long-term impacts of climate change, which exacerbates the frequency and severity of storms and droughts.

Q3. Explain the sources of rainfall in Pakistan and how they vary seasonally. Give examples to support your answer.

The sources of rainfall in Pakistan are primarily influenced by two major weather systems: the monsoon winds and western disturbances. These sources vary seasonally, affecting different parts of the country in distinct ways.

1. Monsoon Winds (Southwest Monsoon)

- **Source:** The monsoon winds originate from the Indian Ocean and the Arabian Sea, bringing moist air towards the Indian subcontinent, including Pakistan.
- **Seasonal Variation:**
 - **Summer (June to September):** The southwest monsoon is the primary source of rainfall during the summer months. It brings heavy rainfall to the eastern and northern regions of Pakistan, particularly in Punjab and Sindh. The northern areas, including the foothills of the Himalayas, receive significant rainfall due to orographic lifting, where moist air is forced to rise over the mountains, cooling and condensing to form rain.
 - **Example:** The city of Lahore often experiences heavy monsoon rains during July and August, which are critical for the Kharif crops like rice and sugarcane.

2. Western Disturbances

- **Source:** Western disturbances are weather systems that originate in the Mediterranean Sea and move eastwards, bringing rain and snow to the western and northern parts of Pakistan.
- **Seasonal Variation:**
 - **Winter (December to March):** Western disturbances are the main source of precipitation during the winter months. They bring rain to the plains of Punjab and Khyber Pakhtunkhwa and snowfall to the northern mountains, including the Himalayas and Hindu Kush ranges.
 - **Example:** Murree and other northern hill stations receive heavy snowfall during the winter months due to western disturbances, which is crucial for the region's water supply in the form of snowmelt during the warmer months.

3. Frontal Rainfall

- **Source:** This type of rainfall occurs when different air masses meet, typically in the transition periods between seasons.
- **Seasonal Variation:**
 - **Spring (March to May) and Autumn (September to November):** During these transitional periods, frontal systems can bring sporadic rainfall to various parts of the country. However, this is not as significant as the monsoon or winter rains.
 - **Example:** Islamabad often experiences brief showers during the spring, which help in moderating the temperature before the onset of the hot summer.

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4. Localized Convective Rainfall

- **Source:** Convective rainfall occurs due to the intense heating of the land surface, causing localized thunderstorms.
- **Seasonal Variation:**
 - **Summer (May to September):** This type of rainfall is more common in the summer months, particularly in the plains and arid regions where the ground heats up quickly.
 - **Example:** The Thar Desert in Sindh occasionally experiences short but intense thunderstorms during the late summer.

Q4. Explain how temperature variations affect the lifestyle of people in different regions of Pakistan.

Temperature variations across different regions of Pakistan significantly impact the lifestyle of its people. Here's how:

1. Northern Mountains (Cold Climate)

- **Temperature:** The northern mountainous regions, including areas like Gilgit-Baltistan, experience harsh winters with temperatures often dropping below freezing, and mild summers.
- **Lifestyle Impact:**
 - **Clothing:** People wear heavy woolen clothing and layers to protect themselves from the cold.
 - **Housing:** Homes are built with thick walls and are often insulated to retain heat. In higher altitudes, houses may have sloped roofs to prevent snow accumulation.
 - **Agriculture:** The growing season is short, limiting agriculture to a few summer months. Crops like wheat, barley, and potatoes are common, and livestock rearing (yaks, goats) is vital.
 - **Transport:** Snowfall often blocks roads in winter, making transport difficult and isolating communities.

2. Indus Plain (Moderate to Hot Climate)

- **Temperature:** The Indus Plain, covering Punjab and Sindh, has extreme temperatures, with scorching summers (often exceeding 40°C) and mild winters.
- **Lifestyle Impact:**
 - **Clothing:** Lightweight and breathable fabrics are worn in summer, while winter clothing is minimal.
 - **Agriculture:** The region is the agricultural heartland of Pakistan, supporting the cultivation of wheat, rice, cotton, and sugarcane. The hot climate allows for multiple cropping seasons.
 - **Housing:** Homes are designed to keep cool, with thick mud walls, high ceilings, and courtyards that provide ventilation.
 - **Daily Routine:** People often adjust their daily routines to avoid the midday heat, with activities planned for early morning or late evening.

3. Coastal Areas (Moderate Climate)

- **Temperature:** Coastal regions, such as Karachi, have a moderate climate with relatively stable temperatures throughout the year due to the influence of the Arabian Sea.
- **Lifestyle Impact:**
 - **Clothing:** Light and breathable clothing is common due to the humid environment.

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- **Economic Activities:** Fishing and port-related activities dominate the economy. The stable climate allows for year-round economic activities.
- **Housing:** Houses are designed to cope with humidity, often featuring large windows for ventilation.

4. Balochistan Plateau (Arid Climate)

- **Temperature:** Balochistan experiences extreme temperature variations, with very hot summers and cold winters, especially in the higher altitudes.
- **Lifestyle Impact:**
 - **Clothing:** People wear layers that can be adjusted for the hot days and cold nights.
 - **Housing:** Homes are made from materials that provide insulation from both heat and cold, such as mud-brick.
 - **Economic Activities:** Livestock rearing and mining are common due to the arid climate, with limited agriculture focused on drought-resistant crops.
 - **Water Usage:** Due to the scarcity of water, conservation is critical, with reliance on karez systems (ancient underground water channels) for irrigation.

5. Desert Regions (Extreme Arid Climate)

- **Temperature:** Deserts like Thar and Cholistan experience extreme heat during the day and cooler nights, with minimal rainfall.
- **Lifestyle Impact:**
 - **Clothing:** Light, loose-fitting clothing is common to protect against the sun while allowing air circulation.
 - **Housing:** Homes are often made from locally available materials like mud and thatch, designed to keep interiors cool.
 - **Economic Activities:** Livelihoods are primarily based on livestock rearing and limited agriculture, often dependent on rainfall or irrigation systems like tube wells.
 - **Water Conservation:** Water is a precious resource, with practices like rainwater harvesting and traditional wells being vital.

✓
Q5. Write short notes on the following topics.

(a) Pressure and Winds

(b) Advantages and Disadvantages of floods in Pakistan.

(a) Pressure and Winds:

Pressure and wind patterns in Pakistan are influenced by the country's varied topography and seasonal changes:

- **Pressure:** The atmospheric pressure in Pakistan varies seasonally, with low pressure dominating during the summer and high pressure during the winter. In summer, the intense heat over the plains, particularly in the Sindh and Punjab regions, creates low-pressure areas, drawing in moist air from the Arabian Sea, leading to monsoon rains. In contrast, winter is characterized by higher pressure, particularly in the northern regions, contributing to drier conditions.
- **Winds:** Pakistan experiences several types of winds:

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- **Monsoon Winds:** These winds bring moist air from the Indian Ocean during the summer, resulting in heavy rainfall in the northern and eastern regions.
- **Western Disturbances:** These are winds originating from the Mediterranean region that bring winter rainfall and snow to the northern and western parts of the country.
- **Local Winds:** In areas like Balochistan, hot and dry winds such as the "Loo" are common during the summer, affecting daily life and agriculture.

(b) Advantages and Disadvantages of Floods in Pakistan:

Floods are a recurrent phenomenon in Pakistan, particularly during the monsoon season. They have both positive and negative impacts:

- **Advantages:**
 - **Soil Fertility:** Floods deposit nutrient-rich silt onto the floodplains, improving soil fertility and supporting agriculture in the subsequent growing season.
 - **Recharging Groundwater:** Floodwaters can help in recharging groundwater levels, especially in arid and semi-arid regions, ensuring water availability during dry periods.
 - **Biodiversity:** Floods can create wetlands and rejuvenate ecosystems, supporting a diverse range of flora and fauna.
- **Disadvantages:**
 - **Destruction of Property and Infrastructure:** Floods cause widespread damage to homes, roads, bridges, and other infrastructure, leading to significant economic losses and displacement of people.
 - **Agricultural Losses:** Crops can be destroyed by floods, leading to food shortages and loss of income for farmers, particularly in regions like the Indus River basin.
 - **Health Risks:** Floods often lead to the spread of waterborne diseases such as cholera and dysentery, as well as an increase in mosquito-borne diseases like malaria and dengue.



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